

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science and Engineering

**CSE-2340: Software Development 1**

# Navigation, Forms and Tables

*Week 2 · Class 1 · Practical HTML patterns*

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# Recall & Today's Outcome

## ? Recall from Week 1

- Which tag groups a set of navigation links?
- How do we connect a label to an input?
- Which four commands push code to GitHub?
- Show me your repository URL.

## ✓ By the end of today you can

- Build a navigation menu that works on every page
- Use the common form controls correctly
- Group related fields with fieldset and legend
- Add basic validation with HTML attributes
- Build a correct data table with headers
- Know when a table is the wrong choice

# Three Patterns on Almost Every Website

Last class you learned individual tags. Today you learn the three patterns that combine them.

## Navigation

How the user moves  
between pages

`nav · ul · li · a`

## Forms

How the user sends  
information to you

`form · fieldset · input  
select · textarea`

## Tables

How you present  
structured data

`table · thead · tbody  
tr · th · td`

**Every one of them is just semantic HTML you already know, arranged in a standard shape**

- Real websites are not new tags. They are the same tags used in patterns that users already recognise.
- Mini Project 1 (Week 3) needs all three. The mid-term practical in Week 7 needs all three.

# Navigation Patterns

```
<!-- Skip link: first thing in the body -->
<a href="#main-content">Skip to main content</a>

<nav aria-label="Main navigation">
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="courses.html">Courses</a></li>
    <li><a href="register.html"
      aria-current="page">Register</a></li>
    <li><a href="#contact">Contact</a></li>
  </ul>
</nav>
```

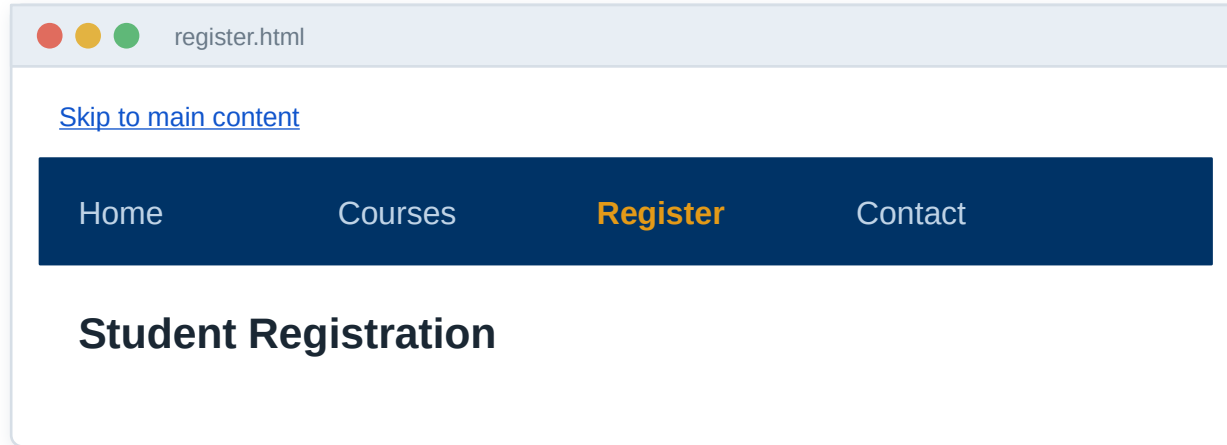
## Three link types

- Page link: `courses.html`
- Anchor link: `#contact`
- External link: `https://...`

## ! Two attributes worth learning

- `aria-label` names the nav when a page has more than one
- `aria-current="page"` marks the page the user is on

# Navigation: What the User Sees



## Rules for a usable menu

- Same menu, same order, on every page
- Mark the current page clearly
- Five to seven items is enough

## Common mistakes

- Putting the links in div tags instead of a ul
- A different menu on each page
- Link text like "click here" or "page 2"
- A menu that only works with a mouse



## Why ul for a menu?

- A menu IS a list of links
- A screen reader announces "list, 4 items"
- The user knows how many choices exist before reading them

# Form Anatomy

```
<form action="#" method="post">

  <fieldset>
    <legend>Personal Information</legend>

    <label for="fullname">Full Name</label>
    <input type="text" id="fullname" name="fullname"
      required>
  </fieldset>

  <button type="submit">Register</button>
</form>
```

**form**

The container. action = where data goes, method = how it is sent

**fieldset**

Groups related fields together

**legend**

The title of that group

**label**

The visible name of one field

**input**

The box the user types in

**button**

Submits the whole form

# Input Types You Will Actually Use

## text

Any short text

## email

Checks for an @

## password

Hides the characters

## number

Digits only, with min and max

## tel

Phone number

## date

Shows a date picker

## url

Website address

## file

Upload a file

```
<label for="email">Email</label>
<input type="email" id="email" name="email" placeholder="you@example.com" required>

<label for="age">Age</label>
<input type="number" id="age" name="age" min="16" max="60">
```

# Radio, Checkbox and Select

```
<!-- RADIO: choose ONE. Same name = one group -->
<input type="radio" id="day" name="shift" value="day">
<label for="day">Day shift</label>

<input type="radio" id="eve" name="shift" value="evening">
<label for="eve">Evening shift</label>

<!-- CHECKBOX: choose MANY. Different names -->
<input type="checkbox" id="html" name="html">
<label for="html">HTML</label>

<!-- SELECT: a dropdown, good for long lists -->
<label for="dept">Department</label>
<select id="dept" name="dept">
  <option value="">-- Choose --</option>
  <option value="cse">CSE</option>
  <option value="eee">EEE</option>
</select>
```



## The radio rule

- Radio buttons only work as a group if they share the SAME name
- Different name = every button can be selected

## When to use which

- 2-4 choices, pick one → radio
- Pick many → checkbox
- 5 or more choices → select

**Label goes AFTER a radio or checkbox, and BEFORE a text input.**

# Validation Without JavaScript

The browser can check simple rules for you. Add attributes, write no code.

**required**

The field cannot be empty

**minlength / maxlength**

Limits how many characters

**min / max**

Limits a number or a date

**pattern**

The value must match a pattern

**placeholder**

A hint. NOT a replacement for a label

```
<label for="sid">Student ID</label>
<input type="text" id="sid" name="sid"
      required
      minlength="8" maxlength="10"
      pattern="[A-Z0-9]+"
      placeholder="C251001">
```

## Important warning

- Browser validation is for convenience only
- A user can bypass it easily
- Real checking must also happen on the server
- That is a later course - but never say a form is "secure" because of required

# Tables: Structure

```
<table>
  <caption>Week 2 class routine</caption>

  <thead>
    <tr>
      <th scope="col">Day</th>
      <th scope="col">Time</th>
      <th scope="col">Topic</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <th scope="row">Saturday</th>
      <td>08:00 - 08:50</td>
      <td>Navigation, forms, tables</td>
    </tr>
    <tr>
      <th scope="row">Sunday</th>
      <td>08:00 - 08:50</td>
      <td>Accessibility essentials</td>
    </tr>
  </tbody>
</table>
```

|                      |                                               |
|----------------------|-----------------------------------------------|
| <b>table</b>         | The whole table                               |
| <b>caption</b>       | Its title. Put it first                       |
| <b>thead / tbody</b> | Header rows / data rows                       |
| <b>tr</b>            | One row                                       |
| <b>th</b>            | A header cell (bold, centred)                 |
| <b>td</b>            | A normal data cell                            |
| <b>scope</b>         | Says whether a th describes a column or a row |

# Tables: Output and Limits

routine.html

*Week 2 class routine*

| Day      | Time          | Topic                     |
|----------|---------------|---------------------------|
| Saturday | 08:00 - 08:50 | Navigation, forms, tables |
| Sunday   | 08:00 - 08:50 | Accessibility essentials  |

## ✓ Use a table for

- A class routine
- Marks or results
- A price list
- Anything with rows AND columns

## Never use a table for

- Page layout - that is what CSS Grid is for (Week 3)
- A list of items with no columns - use ul
- Making text line up nicely

## The test question

- Ask: does my data have a meaning across a row AND down a column?
- Yes → table
- No → it is not tabular data

# 01

## Live Build: Student Registration Page

One page with all three patterns: a nav, a registration form, and a routine table.

# Step 1: Shell and Navigation

```
<body>
  <a href="#main-content">Skip to main content</a>

  <header>
    <h1>IIUC Student Portal</h1>
    <nav aria-label="Main navigation">
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="register.html" aria-current="page">Register</a></li>
        <li><a href="#routine">Routine</a></li>
      </ul>
    </nav>
  </header>

  <main id="main-content">
    <h2>Student Registration</h2>
    <!-- form goes here in Step 2 -->
  </main>
</body>
```

## Step 2: Personal Information Fieldset

```
<form action="#" method="post">

  <fieldset>
    <legend>Personal Information</legend>

    <p>
      <label for="fullname">Full Name</label><br>
      <input type="text" id="fullname" name="fullname" required placeholder="Your full name">
    </p>

    <p>
      <label for="sid">Student ID</label><br>
      <input type="text" id="sid" name="sid" required minlength="7" placeholder="C251001">
    </p>

    <p>
      <label for="email">Email</label><br>
      <input type="email" id="email" name="email" required placeholder="you@example.com">
    </p>

    <p>
      <label for="dob">Date of Birth</label><br>
      <input type="date" id="dob" name="dob">
    </p>
  </fieldset>
```

## Step 3: Choices and Submit

```
<fieldset>
  <legend>Course Preferences</legend>

  <p>Preferred shift:</p>
  <input type="radio" id="morning" name="shift" value="morning" checked>
  <label for="morning">Morning</label>
  <input type="radio" id="evening" name="shift" value="evening">
  <label for="evening">Evening</label>

  <p>Which topics interest you?</p>
  <input type="checkbox" id="html" name="html"><label for="html">HTML</label>
  <input type="checkbox" id="css" name="css"><label for="css">CSS</label>
  <input type="checkbox" id="js" name="js"><label for="js">JavaScript</label>

  <p>
    <label for="dept">Department</label><br>
    <select id="dept" name="dept">
      <option value="">-- Choose your department --</option>
      <option value="cse">Computer Science and Engineering</option>
      <option value="eee">Electrical and Electronic Engineering</option>
    </select>
  </p>
</fieldset>

  <button type="submit">Register</button>
</form>
```

## Step 4: The Routine Table

```
<section id="routine">
  <h2>Class Routine</h2>

  <table>
    <caption>CSE-2340 weekly class routine</caption>
    <thead>
      <tr>
        <th scope="col">Class</th>
        <th scope="col">Day</th>
        <th scope="col">Time</th>
        <th scope="col">Topic</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">C1</th>
        <td>Saturday</td>
        <td>08:00 - 08:50</td>
        <td>Navigation, forms and tables</td>
      </tr>
      <tr>
        <th scope="row">C2</th>
        <td>Saturday</td>
        <td>09:00 - 09:50</td>
        <td>Accessibility essentials</td>
      </tr>
    </tbody>
  </table>
</section>
```

# Common Mistakes and Fixes

## Both radio buttons can be selected

The two inputs have different name values. Give every option in the group the same name.

## Clicking the label does nothing

The for value does not match the input id exactly. Check spelling and capital letters.

## The table looks broken or shifted

A tr, th or td was not closed, or a row has a different number of cells.

## The dropdown submits an empty value

That is the first option with value="". Keep it as the placeholder, and add required.

## The form reloads and clears everything

That is normal. There is no server to receive the data yet.

*Debug method: open DevTools, right click the broken element, choose Inspect, and read the actual HTML the browser built.*

# Assignment - Week 2, Class 1

## 1 Part 1 - In-class demonstration

- Time: last 5 minutes of this class
- Task: add two more rows to the routine table for C3 and C4
- Add one more field to the form: a tel input for a phone number, with a label
- Show that clicking each label focuses its input
- Submission: show on your screen before leaving
- Marks: 2 (checked on the spot)

## 2 Part 2 - Home task

- Build feedback.html: a feedback form with your own fields
- It must contain: two fieldsets, one radio group, one checkbox group, one select, and one textarea
- Add a results table with a caption and scope on every header cell
- Link the new page from your nav on every page
- Commit and push to your GitHub repository
- Deadline: before Week 2, Class 3 · Marks: 3

**Keep register.html. The next class audits and improves it for accessibility.**

48-50 MINUTES

# Marking Rubric & Exit Check

## RUBRIC - 5 MARKS TOTAL

Navigation present and consistent across pages 1.0

Form controls correct: radio group, checkbox, select 1.5

Every input has a correctly linked label 1.0

Table with caption, thead/tbody and scope 1.5

### Exit question 1 - concept

When should you use a table, and when should you use a list?

### Exit question 2 - practical

Write the two lines of HTML for a radio button labelled "Female" in a group named gender.

*Next class: accessibility essentials - we audit and fix the page you just built.*